VisDrone Tracking-by-Detection Report

GitHub Link: https://github.com/jatinkumar3110/Assignment_IISc_Project-Associate

1. Introduction

This work addresses object detection and multi-object tracking in VisDrone aerial imagery, where severe scale variation, frequent occlusion, and dense object layouts challenge both localization and identity continuity.

Aerial viewpoints reduce pixel support for small targets and amplify overlap ambiguity, making false negatives and identity switches more likely than in ground-level benchmarks.

The pipeline follows a tracking-by-detection design: Faster R-CNN generates per-frame detections and a ByteTrack-style IoU association module links them into trajectories.

2. Dataset Description

- The implementation uses VisDrone 2019 detection and MOT validation splits.
- DET-train and DET-val are used for detector training and COCO-style detection evaluation.
- MOT-val sequences are used for sequence inference and tracking evaluation.
- The class space contains 10 VisDrone categories: pedestrian, people, bicycle, car, van, truck, tricycle, awning-tricycle, bus, and motor.

3. Methodology

Detection Model: The detector is Faster R-CNN with a ResNet-50 FPN backbone (`fasterrcnn_resnet50_fpn`) and a classification head adapted to 10 VisDrone classes plus background.

This model is used for both DET validation and frame-wise MOT inference. The choice favors region-based localization quality and multi-scale feature fusion, which are important for crowded aerial scenes with small objects.

Tracking Model: Tracking is a simplified ByteTrack-style, IoU-based implementation.

- Detections are split into high-score and low-score sets.
- Existing tracks are first matched with high-score detections, then unmatched tracks are matched with low-score detections.
- Matching uses IoU with class-consistent association; Hungarian assignment is used when available, with greedy fallback otherwise.
- New tracks are created from confident unmatched detections, and stale tracks are removed after `max_time_lost`.

No appearance modeling (ReID) is used, so identity consistency depends on geometric overlap and detector quality.
This design prioritizes simplicity, interpretability, and reproducibility over maximum robustness under long occlusion.

4. Implementation Details

The notebook is modular and stage-driven, with inline components for dataset handling, preprocessing, detector training/evaluation, tracking inference, and tracking evaluation.

- Stage-based execution is controlled with `PIPELINE_STAGE` and prerequisite expansion.
- Artifacts are persisted under the artifact's directory (checkpoints, summaries, metrics, reports).
- Outputs are consolidated through manifest and publication metadata for reproducibility.

The modular decomposition supports targeted debugging and ablation, while stage-gated execution prevents unnecessary re- computation and enables controlled reruns of specific pipeline blocks.

5. Training Setup

Final run configuration (from `submission_data/run_config.json`):

- Epochs: 4
- Batch size: 2
- Learning rate: 0.005
- Scheduler: StepLR (`step_size=3`, `gamma=0.1`)
- Gradient clipping: `grad_clip_norm=5.0`
- Detection score threshold: 0.30
- MOT score threshold: 0.50
- Association minimum score: 0.40

The 4-epoch schedule reflects practical compute constraints for the end-to-end run.
This budget is sufficient for a stable baseline but limits deeper detector adaptation, which is consistent with moderate localization metrics.

6. Evaluation Metrics

Detection uses COCO-style metrics:

- mAP@0.50 and mAP@0.50:0.95 measure localization and classification quality at single and multi-IoU settings.
- mAR@0.50 and mAR@0.50:0.95 measure recall capacity across IoU settings.
- Precision and Recall summarize false-positive and miss behavior.

Tracking uses MOT metrics:

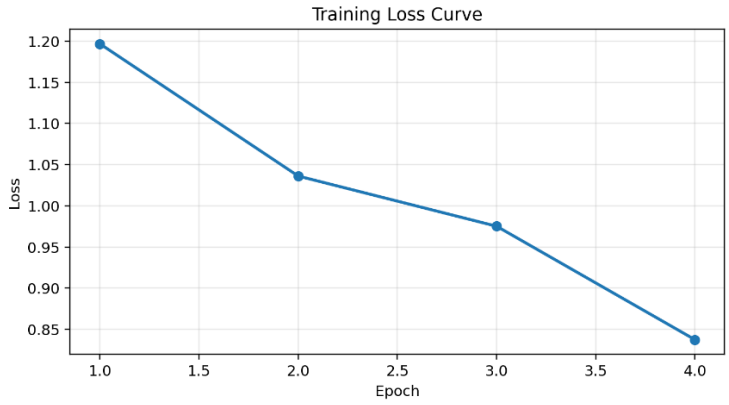
- MOTA summarizes overall tracking accuracy under FP, FN, and ID switch penalties.
- HOTA measures balanced detection-association quality.
- IDSW counts identity switches across frames.

7. Results

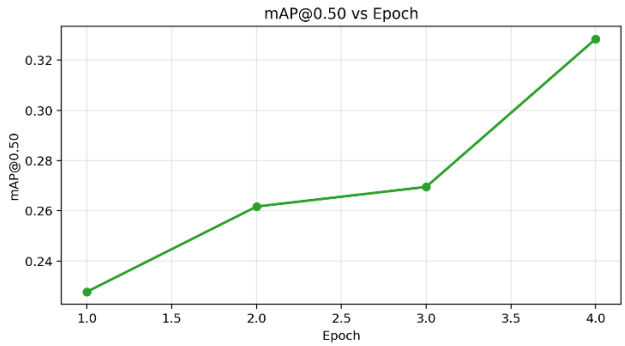
Final values (from `submission_data/metrics_summary.json`):

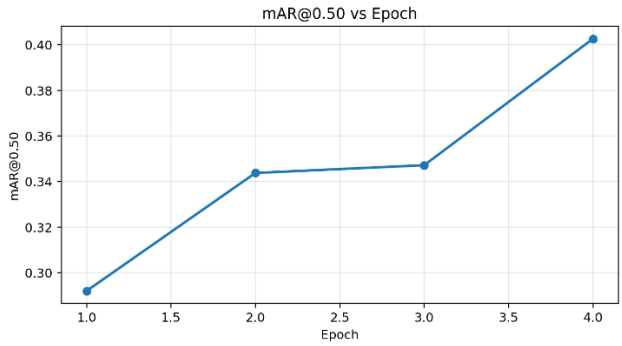
Detection	Tracking (overall)
• mAP@0.50:0.95 = 0.1964	• MOTA = 0.1157
• mAP@0.50 = 0.3285	• HOTA = 0.3641
• mAR@0.50 = 0.4026	• IDSW = 264
• mAR@0.50:0.95 = 0.2603	• ----------
• Precision = 0.1964	• Precision = 0.6194
• Recall = 0.2603	• Recall = 0.3062

- mAP and mAR together indicate a detector that covers a meaningful fraction of objects but remains limited on strict localization quality.
- The recall-above-precision pattern indicates broader candidate coverage at the cost of additional false positives, a common regime in dense aerial scenes.
- Tracking combines low MOTA with moderate HOTA, suggesting that identity and miss-related penalties remain high even though association quality is non-trivial for matched detections.



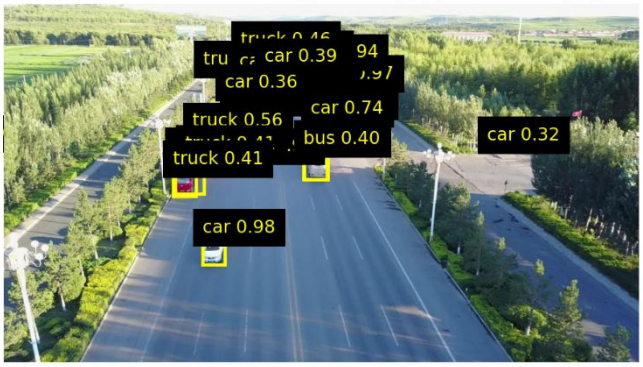
Training loss trend across epochs, indicating optimization behavior during detector fine-tuning.



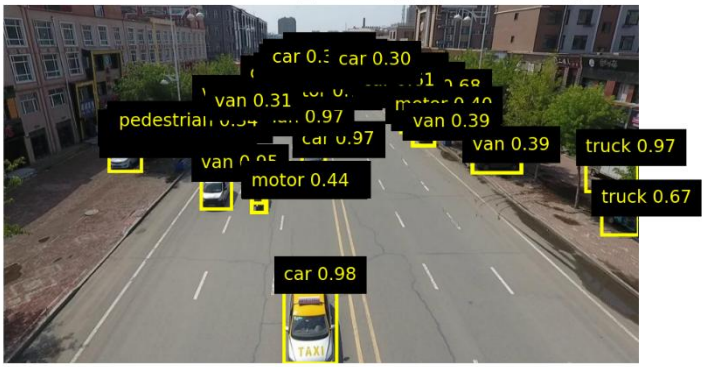


Epoch-wise validation Map and mAR progression for detector quality monitoring
and highlighting recall behavior respectively.

Detection: 0000276_02601_d_0000520.jpg

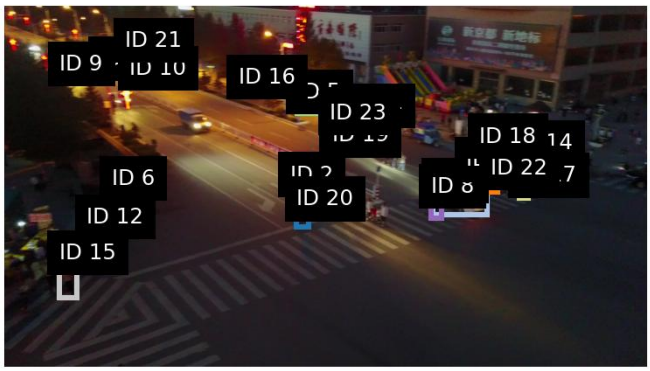


Detection: 0000242_00627_d_0000003.jpg



Qualitative detection examples on validation images, including dense and small-object scenes.

uav0000117_02622_v | frame 1



uav0000086_00000_v | frame 1

Qualitative tracking outputs showing trajectory continuity and identity consistency.

8. Observations

- The detector exhibits recall-biased behavior, capturing a wider object set at the expense of higher false-positive load.
- Tracking precision significantly exceeds tracking recall, indicating conservative temporal linking after score and IoU filtering.
- Upstream detector misses directly reduce trajectory continuity, amplifying fragmentation and lowering MOTA.
- Small-object scale and crowd density remain the dominant coupled failure mode for both detection and identity persistence.

9. Limitations

- The absence of ReID contributes directly to elevated IDSW by weakening identity recovery after occlusion.
- IoU-only association is brittle under abrupt motion, scale change, and prolonged overlap, reducing identity consistency.
- Limited training duration (4 epochs) likely contributes to moderate mAP/mAR rather than stronger localization performance.
- Dense aerial scenes with tiny targets remain difficult for both precise box regression and long-horizon track stability.

10. Future Improvements

- Add appearance-based ReID to improve re-association after occlusion and reduce ID switches.
- Upgrade the detector to architectures with stronger small-object sensitivity to raise recall without disproportionate false positives.
- Increase training duration and perform targeted hyperparameter tuning to improve localization fidelity.
- Add richer motion modeling to stabilize association under camera motion and non-linear object trajectories.

11. Conclusion

The final system is a reproducible baseline for aerial tracking-by-detection built on Faster R-CNN and simplified ByteTrack-style IoU association.

Its results expose a clear operating profile: reasonable coverage and association precision, but limited localization strength and identity persistence under dense occlusion. This makes the improvement path explicit and technically grounded: stronger detection, appearance-aware association, and richer temporal modeling.